Integration Layer & Service Topology

Expanding our Marketplace Services across VZ

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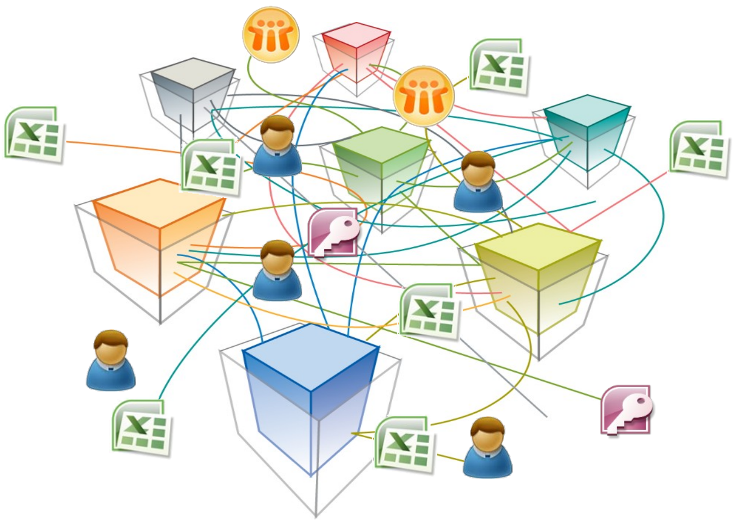
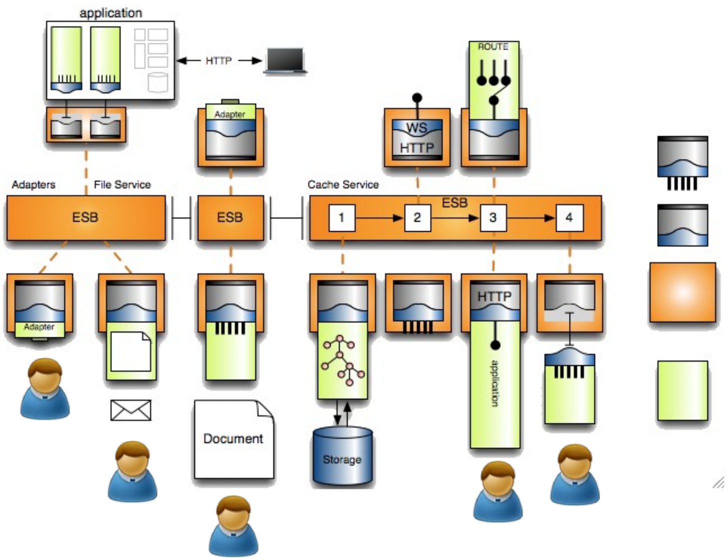
Short List of Outcomes

- Reduce the likelihood of issues cascading across systems and/or components
- Reduce the time required to diagnose and resolve issues across the Digital Supply Chain (1VZ Ecosystem)
- Reduce the number of components to manage
- Reduce the total cost of operations for Digital Supply Chain
- Reduce the time required to extend existing processes
- Reduce the amount of human intervention required to support Digital Supply Chain
- Increase the likelihood of extending services beyond Digital Supply Chain to consumers across VZ
- Reduce the need to test components when changes are introduced into Digital Supply Chain
- Increase the amount of reuse of tools and technologies
- Reduce the amount of duplicate functionality across the Digital Supply Chain
- Increase the likelihood that consumers can search & find data assets across VZ
- Reduce the amount of effort and investment required to build and operate DTs
- Reduce the need to modify legacy systems and solutions that wish to leverage Data, Metrics, & Models
- Reduce the likelihood of business downtime when change is introduced
- Reduce the effort and time to introduce and extend DAG services for D&A customers

Big Picture





Managing Change & Complexity at Scale

Short List of Use Cases (not exhaustive):

- Create a plug-n-play capability between all Maap components and tools to eliminate point-to-point direct connections to improve abstraction and resiliency
- Insulate all Digital Supply Chain components and tools from changes to data, systems, and platform
- Reduce vendor lock-in by insulating tools from solutions for D&A services.
- Simplify the implementation for POCs and swap-outs for tools and components
- Reduce the impact of changes from patches and upgrades to platforms, systems, and components in the Digital Supply Chain . This includes pipelines that feed the GCP as well as delivery tools and pipes used to feed the Data Consumption Layer (SPOG, AI, BI)
- Create a set of reusable services from the function points and components within Digital Supply Chain to support current needs and improve extensibility in the future
- Create Search as an enterprise utility that can be used for finding Data (initially) with progressive capabilities to support semantic search and answer questions in the future

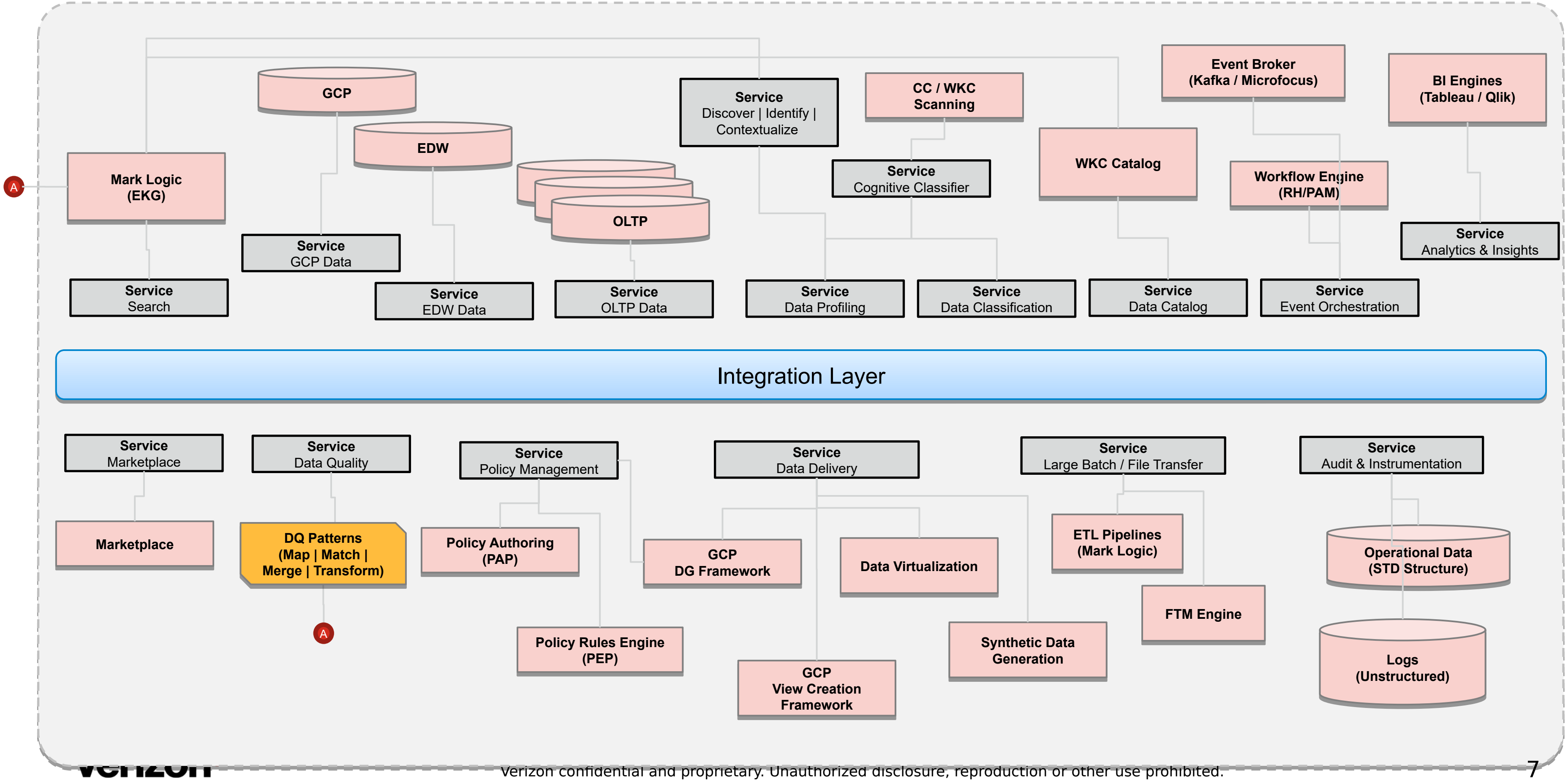
Short List of Use Cases (not exhaustive):

- Extend reusable services for VZ consumers (systems, components) as a set of consumables they can subscribe to and/or incorporate (via our DAG integration layer or individually if they wish) covering but not limited to:
  - Data Profiling
  - Data Quality
  - Data Contextualization
  - Data Classification
  - Data Enrichment
  - Data Publishing (via Catalog)
  - Search (as an evolving enterprise utility)
  - Persona-Driven Experience (search, SPoG, notification)
  - Data Provision (ADOM Groups => SSO as a service offered by CISO)
  - Data Protection as a Service
  - Data Integration (extending plug-n-play across VZ)
    - covers batch, near-time, real-time
    - covers episodic, scheduled, and event-driven triggers
  - Process Integration
    - patterns are similar to data but there are different pattern, triggers, and flows
  - Fulfillment (via Marketplace) an Delivery

Short List of Use Cases (not exhaustive):

- Enable any service owner to create their own portal using a suite of tools
- Enable any service owner to create their own Dashboard using a suite of widgets and visualization components
- Extend the capabilities for DCAM, DG, Compliance (security, privacy, protection) to support 1VZ ecosystems
- Set the foundation to support Citizen Developers so they can build apps that can answer complex questions (queries)
- Last but not least we want to extend Monitoring & Measurement as a suite of services that can leverage the DAG services & components:
  - Data Onboarding so their logs and data can leverage our Maap supply chain
  - Data Classification & Data Protection
  - Data Models (algorithms)
  - Data Measures (analytics)
  - Dashboards and Alerts

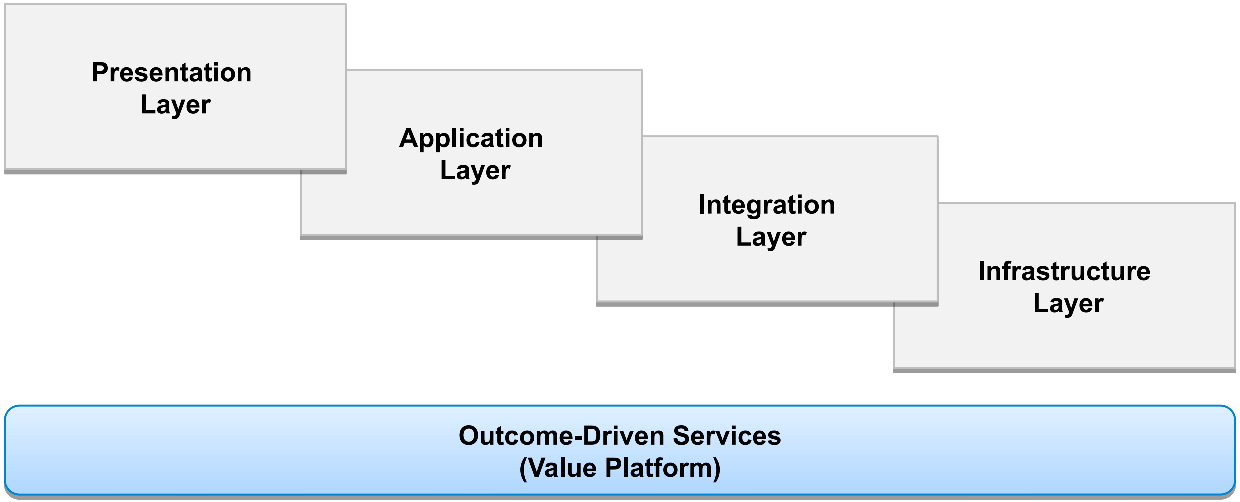
Conceptual View of an Integration Layer



Towards N-Tier Design

The Integration Layer provides several benefits (insulation, abstraction, resiliency, scalability, managed complexity, avoids vendor lock-in, and…

- Provides a path towards introducing N-Tier architecture for the Marketplace Service Topology
- Creates a natural way to introduce services and introduce the components & tools into logical containers to simplify architecture for accelerating Design Time & Run Time service delivery



N-Tier: Goal & Purpose

The goal is to create areas of separation between the architecture layers, simplify the process of introducing change, and improving overall agility. We'll focus on the following 3 tiers:

- Presentation Layer
- Application Layer
- Integration Layer

Infrastructure Services
- Compute & Memory
- Storage Management
- Core Engines & Gateways
- Messaging & Networking
- Enterprise Utilities

The purpose is to create functional separation and control within the appropriate layer of the architecture. Once the ARB agrees on these standards, we (the ARB) will begin introducing design enhancements w/in future product/system releases on a per project basis.

N-Tier: Presentation Layer (standards)

- HTML5 going forward
- Browser / Device Indep.
- Light Validation only (tbd)
- Navigation
- No calls to the data persistence layer
- Fields all pulled from Data Dictionary
- Labels, Formats, and Display characteristics pulled from DAL *
- No transforms, data manipulation or translations (performed by DM)
- Active Profiling of Application
- Active Profiling of Users
- Shift from applications to veneer
- Component-Based Design (Web 2.0 -> 3.0)

These are only a few of the standards for defining what should be included or excluded in the presentation layer

*** Data Access Layer**

N-Tier: Application Layer (standards)

- Event Driven
- Event Triggers are well defined
- Logic is managed centrally
- Status Rules are embedded during Design-Time
- Variable Rules are provisioned during Run-Time
- Functional Execution is controlled by Logic
- Process Behavior is controlled by Logic
- Security is managed by Logic
- Data is managed by Data Management Framework (via DAL)
- Services represent a finite (limited) # of function points
- Large segments of code with embedded logic should be re-factored
- COTS & 3rd Party Engines should be leveraged where possible

These are only a few of the standards for defining what should be included or excluded in the application layer

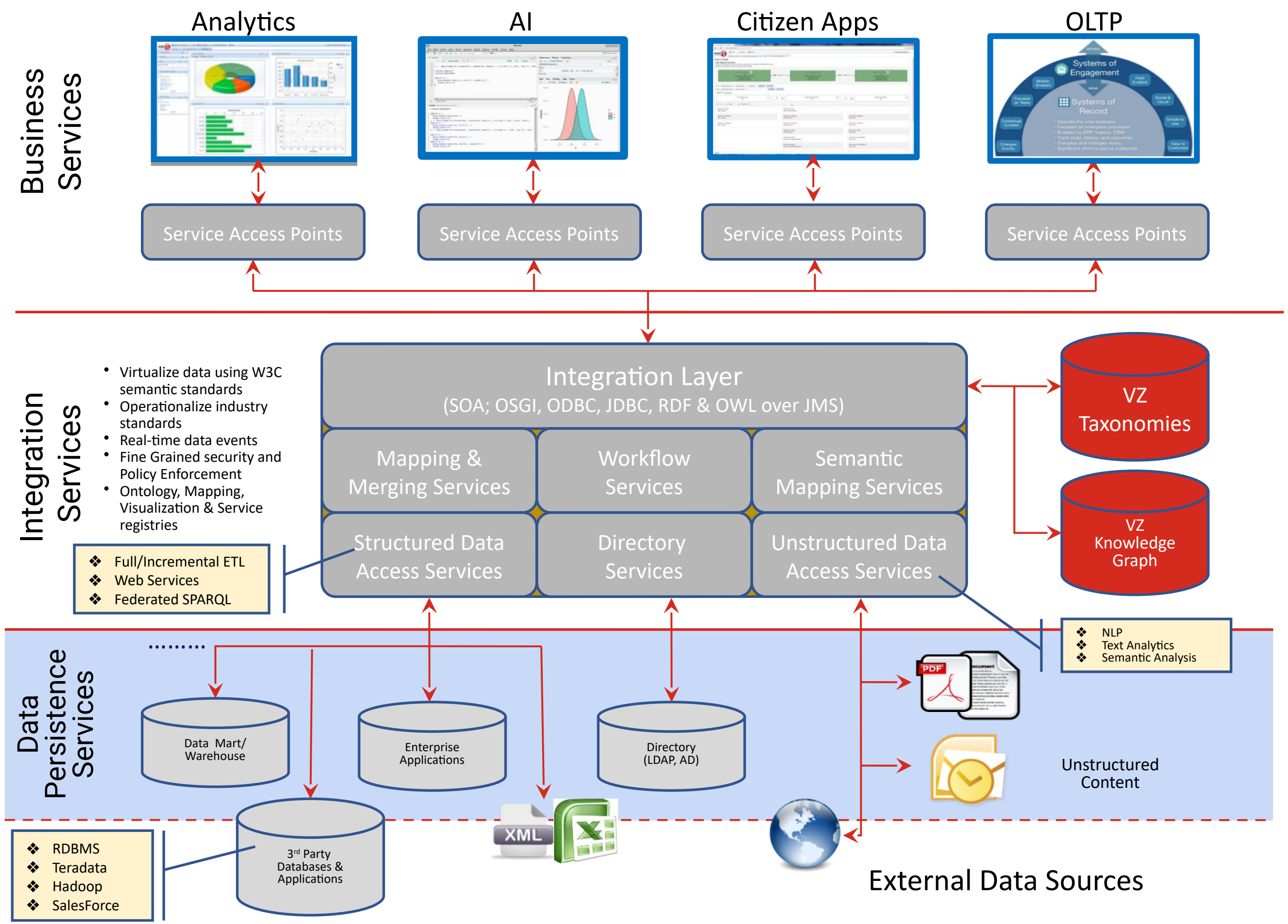
N-Tier: Integration Layer (standards)

- Contains one or more engines (part of a suite)
- Contains a set of utility services
  - Data access services
  - Data integration services
  - Data acquisition services
  - Data extraction services
  - Data warehousing services
  - Data profiling, cleansing, & measurement services
  - Data transformation, translation, and transliteration services
  - Data integration services (between Arch. Tiers)
- May contain a number of ODSs and/or repositories for staging data

These are only a few of the standards for defining what should be included or excluded in the application layer

Note: Data Persistence is the Physical instantiation of logical data partitions and structures.

Future State (3-Tier Architecture)



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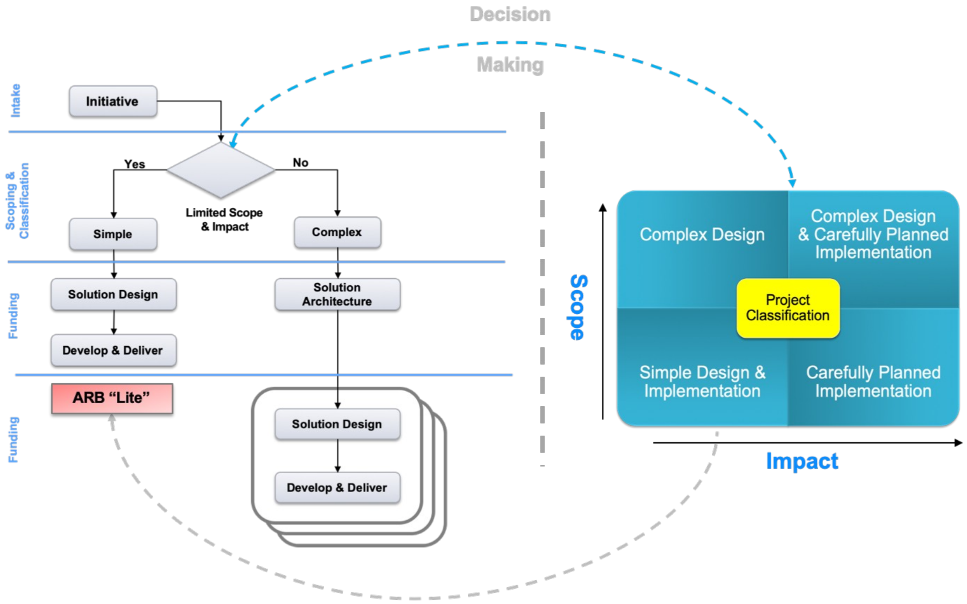
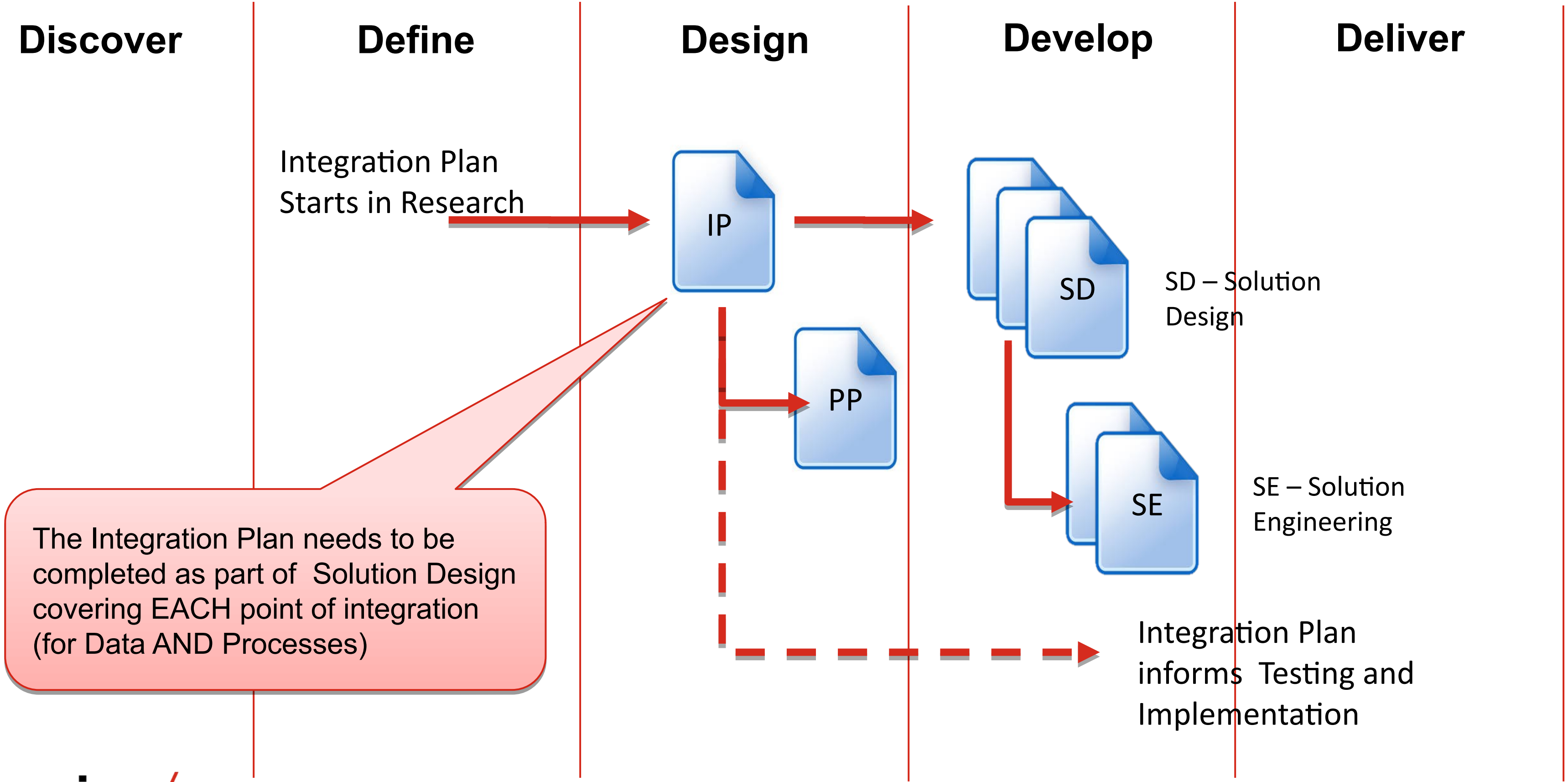
Integration Planning (for EVERY Component / Tool)

1. Guiding Principles
2. General Information
3. Integration Architecture Drawing
4. Endpoint Evaluation
5. Endpoint Requirements
6. Process and Data Flow
7. Canonical Models
8. Provider Services
9. Consumer Services
10. Cost and Time Estimates
11. Summary



Integration Planning informs Design & Develop & Delivery



Integrate vs Interoperate

Integrate

- Process of combining multiple applications to function together as a unified whole.
- A more integrated application environment with proprietary methods of connectivity or data sharing can slow the proliferation of the many innovative, disruptive point technologies we see produced from new vendors without a clear and easy path to connect them with other systems.

Interoperate

- More advanced and meaningful than integration.
- Incorporates content from multiple disparate and entirely independent systems to advance the effective delivery of solutions to the market
- Bi-directionally sharing data and information
- All tools and components speaking the same language
- All tools and components conform to the same standards of interoperability
- Promotes TRUST by assuring that these systems can work together simply and reliably
- Ensures faster and lower-cost implementations
- Supports more predictable methods of implementation with multiple vendors, and shortens the time and decreases the cost of initial ownership for the overall solutions being deployed
- Standards in interoperability can be used to qualify or certify conforming partners so we can deploy new interoperable solutions with little to no impact on development resources.

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E2E Service Orchestration for 1VZ Data

Classify | Catalog (Data & Metrics) | Governance Services | 3rd Party Sharing | Monitor & Measure Controls

Sourcing Services
- Acquire
- Procure
- Ingest
- Protect
- Inventory
- Quarantine
- Qualify
- Stage
- Warehouse

Curation Services
- Transform
- Merge
- Assemble
- Map
- Harmonize
- Attribute
- Optimize
- Semantic Enrichment
- Protect
- Master
- Persist
- Profile
- Index
- Cleanse
- Publish
- Consolidate

Analytic & Insight Services
- Metrics Design
- Dimension Design
- Semantic Enrichment
- Causal Mapping
- Catalog
- Aggregate
- Validation
- Projection
- Reason
- Explain
- Publish
- Monitor
- Visualize

Consumption
- Connection Services
- Bundle & Abstraction
- Search (Basic > Semantic)
- Publish
- Monitor

Experience
- Provision
- Personalize
- Search
- Explore
- Tailor
- Configure
- Visualize
- Publish
- Explain

Integrate | Orchestrate | Schedule | Lineage & Reproducibility | Change Control | Configuration | Release Management

Triage | Diagnose | Support | Operational Services | Communicate | Coordinate | Monitor & Measure

Compute | Storage | Environment | Infrastructure Services | Network | Utilities

Agenda for Workshop (Day 1)

2 Hours:

- **Align on the Desired Outcomes (slide 2)**
- **Align on a few Use Cases (directionality) (slides 4-6)**
- **Align on Architecture Principles driving Integration (slides 7-9, 16)**
- **Align on Design Standards for integration (Data & Process) (slides 10-12, 14-15)**
- **Align on the Scope for the Digital Supply Chain (slide 17)**

2 -4 Hours:

- **Discuss Event Driven Architecture (EDA) Topology**
- **Discuss Workflow, Orchestration, and Choreography (and handoffs between EDA)**
- **Discuss Design Standards covering Integration as a Service (IaaS)**

Agenda for Workshop (Day 2)

2 Hours:

- **Socialize Integration Patterns that need to be stood up**
- **Discuss the need for Design Reviews and Controls with each integration (service)**
- **Discuss Operations including Monitoring & Measurement for Integration Services**
- **Discuss need for COE to guide teams (away from P2P towards Events & Topics)**
- **Share Best Practice Design examples of Integration Services**

2 Hours:

- **Breakout: DAG Integration Services & Tool Wrappers**
- **Breakout: 1Corp Integration Services & Tool Wrappers**
- **Breakout: 1Network Integration Services & Tool Wrappers**

2 Hours:

- **Readouts by Teams (share Target State for Integration)**
- **Roadmap of Epics required to support Target States**